

Knowledge-Enhanced Multimodal Fake News Detection: Semantic Visual and Priority Fusion

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Abstract

Multimodal fake information increasingly threatens the Web ecosystem's trustworthiness and security, making improving detection accuracy a critical scientific challenge. The limited information interaction in traditional multimodal fake news detection methods fails to leverage semantic knowledge to model complex cross-modal forgery patterns and global structural anomalies, restricting the model's capability. To address the issues, this paper proposes a multimodal fake news detection method, SVPF-Net, that centers on semantic-driven visual enhancement and knowledge-aided modality-priority fusion. For visual representation optimization, we design a dual-feature extraction module and a dual-fusion enhancement module. A weighted fusion strategy is employed to construct a structured visual representation that integrates the semantics of local forgeries and global anomalies. Meanwhile, a cross-attention mechanism enables bidirectional alignment and interactive coupling between local and global image features, thereby achieving effective complementarity between local forgery cues and global anomaly patterns. For multimodal fusion, high-quality textual semantic features and visual representations are integrated via a modality-priority progressive fusion strategy that relies on cross-attention. The integration enables robust cross-modal semantic interaction and effectively enhances the efficiency of multimodal feature fusion. Comprehensive experiments validate the optimal performance of SVPF-Net and its ability to enhance interpretable semantics, providing valuable support for the practical application of reliable fake news detection.

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WWW '26, Dubai, United Arab Emirates
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ACM ISBN 979-8-4007-2307-0/2026/04
<https://doi.org/10.1145/3774904.3792729>

CCS Concepts

• **Information systems** → **Data mining**; *Information retrieval*; • **Computing methodologies** → **Natural language processing**; *Computer vision*; • **Human-centered computing** → *Social media*.

Keywords

Fake news detection, Multimodal fusion, Cross-attention mechanism, Semantic visual representation

ACM Reference Format:

Qin Zhang, Jiaying Liu, Qian Tao, Zhiwei Guo, Qiyue Zhong, Yifan Zhang, and Ziyan Huang. 2026. Knowledge-Enhanced Multimodal Fake News Detection: Semantic Visual and Priority Fusion. In *Proceedings of the ACM Web Conference 2026 (WWW '26)*, April 13–17, 2026, Dubai, United Arab Emirates. ACM, New York, NY, USA, 10 pages. <https://doi.org/10.1145/3774904.3792729>

1 Introduction

The Web has evolved into the core ecosystem for global information exchange and value dissemination. In the era of all-media communication, fake news proliferates across social platforms, evolving from purely textual forms to intricate combinations of text, images, and videos. The rapid spread of multimodal fake information severely undermines user trust and poses a fundamental threat to downstream applications that rely on structured, trustworthy data. In this context, developing accurate detection models for multimodal fake information in complex network environments has become a critical research priority to ensure the trustworthiness and resilience of the Web ecosystem [31, 43, 48]. Recent advances in artificial intelligence, particularly deep learning, have substantially improved multimodal fake news detection performance, with integrating textual, visual, and sometimes auditory information [38]. These approaches typically leverage feature-level, decision-level, or hybrid fusion mechanisms to improve classification accuracy [16, 44].

However, existing multimodal fake news detection methods suffer from two core limitations that are closely intertwined with

semantic understanding and knowledge integration. First, they fail to effectively capture and associate two complementary, interpretable semantic cues in fake images—local forgery traces and global structural anomalies, due to the fragmented extraction of local and global visual features [17, 42]. Although Vision Transformer (ViT) [8, 11] can capture global structural semantics but often loses fine-grained local details. Second, traditional multimodal fusion frameworks often treat text and vision as equal or rely on simple concatenation. This limited interaction fails to leverage semantic knowledge for modeling complex cross-modal forgery patterns, ultimately hindering the comprehensive detection of fake content.

To address the aforementioned issues, this paper proposes a knowledge-enhanced multimodal fake news detection method, SVPF-Net, based on semantic visual and priority fusion. With semantic knowledge as the core driver, SVPF-Net improves both the interpretability and discriminative power of fake information detection from two key perspectives: visual semantic enhancement and cross-modal knowledge alignment. Specifically, in the text modal feature extraction module, the latest pre-trained language model, RoBERTa is adopted to endow text features with stronger semantic consistency and knowledge interpretability. Second, the visual feature extraction module incorporates a semantic enhancement framework. This design strengthens the capture of interpretable cues, encompassing both local forgery traces and global structural anomalies within the fake images. Furthermore, to solve the problems of low interaction efficiency between modalities and unbalanced semantic fusion, this paper also designs a knowledge-guided modality-priority progressive fusion strategy based on co-attention. This strategy enables selective integration of cross-modal information, ensuring robust semantic interaction while mitigating fusion imbalance. The main contributions of this paper are summarized as follows:

- We propose the semantic visual enhancement module, which integrates the local perception of Convolutional Neural Networks (CNNs) with the global structural modeling of Vision Transformers (ViTs).
- We propose a knowledge-guided priority fusion strategy that assesses the relative importance of each modality. By employing a collaborative attention mechanism, this strategy establishes semantic dominance and facilitates progressive fusion, ultimately enhancing the model's capacity for cross-modal knowledge interaction and reasoning.
- We conduct extensive experiments on publicly available real-world social media datasets, demonstrating that SVPF-Net significantly outperforms state-of-the-art methods in multimodal fake news detection.

2 Related Work

2.1 Single modal fake news detection

Supervised fake news detection methods based on text take the textual information of news as input to identify fake news [29, 30]. Early approaches relied on distributed representations such as Word2Vec [28] and shallow neural networks like CNN [21] and RNN [27]. Although these methods could extract local semantic patterns and sequence dependencies, they were limited by the defect

of fragmented contextual understanding [2, 13, 23, 50]. The emergence of pre-trained language models has achieved a technological breakthrough. BERT [7] improved global semantic representation through large-scale self-supervised learning, while RoBERTa further enhanced the ability to capture semantic details by optimizing pre-training strategies, making it a preferred model for extracting textual fake features. For fake information detection in the image modality, the focus lies in identifying tampering traces and semantic anomalies [24, 33]. In the stage of CNN-dominated local feature extraction, CNN became the mainstream due to its strong local feature learning capability [22]. VGGNet [35] deepened the feature hierarchy by stacking small convolutional kernels, and ResNet [12] addressed the gradient vanishing problem in deep networks using residual connections. Inspired by the NLP field, ViT [8] segments images into token sequences and realizes efficient global feature modeling via the self-attention mechanism, demonstrating stronger generalization ability in deepfake detection. However, ViT is less capable than CNN in capturing subtle local tampering traces, suffering from the defect of loss of local detail. In the exploration stage of spatial-frequency domain feature fusion, researchers have begun to pay attention to the supplementary value of frequency-domain information. Qi et al. [33] proposed MVNN, which fuses frequency-domain tampering traces captured by CNN with semantic features in the pixel domain. Meanwhile, some studies have attempted to combine the advantages of CNN and Transformer. Huang et al. [15] proposed ViT-AMCNet, which integrates ViT with a convolutional module embedded with an attention mechanism. Efficient ViT directly adopts the architecture of CNN feature extraction combined with Transformer encoding, initially realizing the synergy between local and global features. However, these methods are mostly applied to general image classification tasks and lack specialized optimization for manipulating features targeted at detecting fake news.

2.2 Multi-modal Fake News Detection

The landscape of misinformation detection has evolved from rudimentary unimodal analysis to a more sophisticated stage of multi-modal fusion [1, 9, 19, 47]. Gao et al. [10] put forward a dynamic multi-modal feature fusion method that enables the alternating transfer of dynamic information between visual and linguistic modalities. Huang et al. [14] proposed a deep multi-modal attention fusion model (DMAF) that exploits distinguishing features and inherent correlations between visual and semantic content. SpotFake [37], simple yet effective, independently transforms features from text and images and finally uses a fully connected neural network to classify the news as real or fake. The same authors later proposed SpotFake+, which incorporates multi-modal semantic information and transfer learning to capture the semantic and contextual information present in news texts and their corresponding images for better fake news detection accuracy [36]. Wu et al. [42] further stacked multiple co-attention layers to fuse spatial-domain, frequency-domain, and text features, enhancing the model's capability to learn interdependencies across features. The HMCAN model [34] combines hierarchical semantics with the Co-attention Mechanism (CA) to capture rich linguistic information. Choi et

al. [5] proposed a multi-modal feature fusion method for imbalanced datasets to achieve sentiment recognition by comparing and investigating different feature fusion approaches for text and speech. Chen et al. [3] developed a multi-modal rumor detection network for social rumor detection by integrating complementary information from text and images. Wang et al. [40] proposed the MSACA network, which addresses inter-modal semantic misalignment via multi-scale semantic alignment and adaptively selects the most discriminative features by integrating scale-wise cross-modal attention. Considering the prevalence of intramodal noise and intermodal quality discrepancies in real-world data, Lv et al. [26] proposed the dynamic fusion framework MDF-FND, integrating uncertainty quantification and Dempster-Shafer (D-S) evidence theory into the multimodal fusion decision-making process.

However, the visual modality in existing studies lacks dedicated fusion of local and global features, and the fusion strategies also overlook the dynamic differences in the importance of modalities. These shortcomings make it difficult for the models to fully capture cross-modal collaborative forgery features, limiting the further improvement of detection accuracy and robustness.

3 Method

We propose a multimodal fake news detection method, namely SVPF-Net, based on enhanced visual representation and modality priority progressive fusion. The overall framework of the model is illustrated in Figure 1. The model consists of three main stages: (1) A **Feature Extraction Module** that uses RoBERTa for textual features (Ω_T), a CNN for frequency-domain features (Ω_F), and our proposed FND-Conformer for visual-spatial features (Ω_V). (2) A **Feature Fusion Module** that employs a progressive, co-attention-based strategy, first fusing text and visual-spatial features ($\Omega_C^{(1)}$), and then integrating frequency features ($\Omega_C^{(2)}$). (3) A **Fake News Detection Module** that uses the final fused representation (Ω_{Final}) for classification. The bottom panel details the **FND-Conformer**, which leverages a dual-branch architecture to capture both local (CNN branch) and global (Transformer branch) visual information.

3.1 Feature Extraction

3.1.1 Text modal feature extraction module. In terms of text modal feature extraction, we use the pre-trained language model RoBERTa as the basic feature extractor. RoBERTa has demonstrated excellent performance in natural language processing tasks such as text classification and semantic understanding by optimizing the pre training strategy of BERT and learning more general language knowledge and deep semantic representation ability on large-scale unlabeled text corpora [25, 30].

The text contents are first performed standardized preprocessing to obtain a text sequence that can be input into the model. The preprocessed text content is represented as a word sequence $T = [T_1, T_2, \dots, T_n]$, where n represents the number of words in the tweet. Each word $T_i \in T$ is segmented using a pre-trained vocabulary, as shown below:

$$H_1 = \text{RoBERTa}([T_1, T_2, \dots, T_n]) \quad (1)$$

Among them, the RoBERTa model maps the input sequence to a hidden state matrix $H_1 \in \mathbb{R}^{k \times h}$, where k is the sequence length and h is the dimension of the hidden layer.

On the basis of the output of RoBERTa, a fully connected layer and an activation layer are connected. The mathematical expression is:

$$\Omega_T = \text{ReLU}(W_T \cdot \text{Pooling}(H_1) + b_T) \quad (2)$$

where matrix H_1 is compressed into a vector representation through pooling operation, and the fully connected layer parameters $W_T \in \mathbb{R}^{d \times h}$ and bias vector $b_T \in \mathbb{R}^d$ perform dimension transformation on the features, resulting in the final text feature $\Omega_T \in \mathbb{R}^{d \times 1}$.

3.1.2 Semantic visual enhancement module. To effectively capture both the subtle, local artifacts and the broader, structural inconsistencies in fake images, we propose the **FND-Conformer** based on Conformer[32]. This module synergistically combines the strengths of CNNs for local feature extraction and ViTs for global context modeling. As depicted in the diagram, the FND-Conformer primarily comprises two parts: a Dual Information Extraction Module and a Dual Fusion Enhancement Module.

First, the input image $I \in \mathbb{R}^{H \times W \times 3}$ ($H = W = 224$) is pre-processed to extract low-dimensional features. This stage uses a large-kernel convolution and max-pooling to reduce spatial dimensions while preserving key features, yielding a feature map Y for the subsequent branches.

$$Y = F_{\text{Maxpooling}3 \times 3}(\sigma(L_{BN}(F_{\text{Conv}7 \times 7}(I)))) \quad (3)$$

Dual Information Extraction Module. This module leverages two parallel branches for comprehensive feature extraction:

1) **Local Feature Extraction Branch.** This branch employs a ResNet-like CNN structure to capture fine-grained local details. Its hierarchical nature is adept at identifying subtle tampering evidence, such as edge artifacts or unnatural textures. The feature update for the l -th layer is:

$$\mathbf{x}_L^{l+1} = \mathbf{x}_L^{l'} + \mathcal{F}_{\text{CNN}_l}(\mathbf{x}_L^l) \quad (4)$$

2) **Global Representation Extraction Branch.** Concurrently, a ViT branch captures global structural information and long-range dependencies, which is crucial for identifying semantic inconsistencies. The input map Y is partitioned into patch embeddings, $\mathbf{z}_G^0 = F_{\text{PatchEmbed}}(Y)$, which are then processed by standard Transformer encoder blocks:

$$\mathbf{z}_G^l = \text{MHSA}(\text{LN}(\mathbf{z}_G^l)) + \mathbf{z}_G^l \quad (5)$$

$$\mathbf{z}_G^{l+1} = \text{MLP}(\text{LN}(\mathbf{z}_G^l)) + \mathbf{z}_G^l \quad (6)$$

Dual Fusion Enhancement Module. To integrate the extracted features, this module utilizes a series of interactive FND-Conformer blocks. These blocks serve as fusion units where the two branches mutually enhance one another. Specifically, a feature coupling mechanism is employed to facilitate cross-branch interaction, where local details are injected into global representations via downsampling, while local features are simultaneously refined using global context through upsampling.

After passing through these fusion blocks, the final representations from both branches are combined into a unified visual feature vector Ω_V using a learnable weighted sum:

$$\Omega_V = \alpha \cdot \text{GlobalAvgPool}(\mathbf{x}_L^{\text{final}}) + (1 - \alpha) \cdot \mathbf{z}_G^{\text{final}}[0] \quad (7)$$

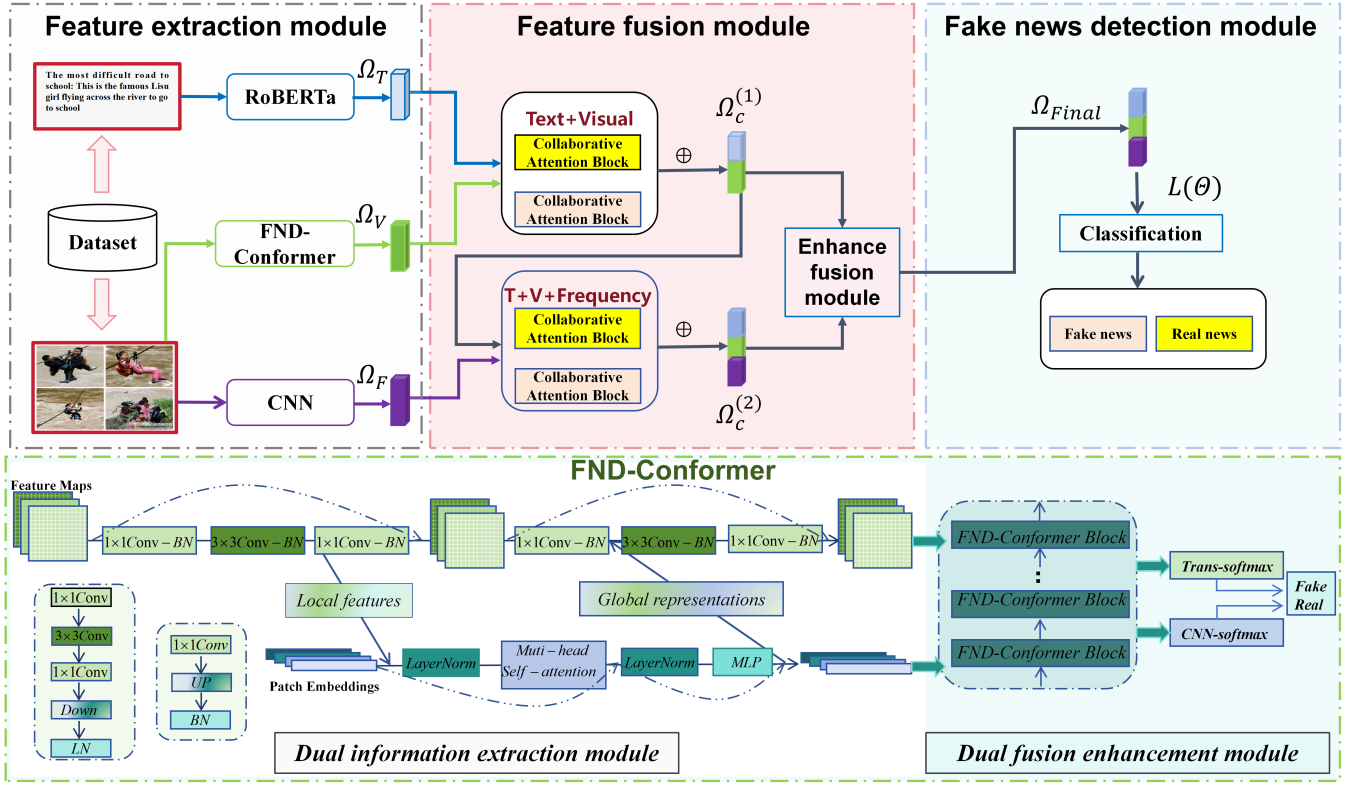


Figure 1: The overall architecture of our proposed SVPF-Net model.

Here, x_L^{final} represents the refined feature map from the CNN, $z_G^{final}[0]$ is the ViT's class token, and α is a learnable weight. This enhanced feature Ω_V provides a comprehensive visual representation for subsequent fusion with other modalities.

3.1.3 Image frequency domain modal feature extraction module. The frequency domain features of images in false information also contain rich information, especially in detecting whether images have undergone specific processing, which has important value. We use the discrete cosine transform to transform the original image from the spatial domain to the frequency domain, with the following formula:

$$F(u, v) = \alpha(u)\alpha(v) \sum_{x=0}^{M-1} \sum_{y=0}^{N-1} f_I(x, y) \cos\left(\frac{(2x+1)u\pi}{2M_I}\right) \cos\left(\frac{(2y+1)v\pi}{2N_I}\right) \quad (8)$$

where $f_I(x, y)$ represents the image in the spatial domain, and after DCT transformation, the frequency-domain image $F(u, v)$ is obtained. M_I and N_I correspond to the size of the image in the x and y directions, respectively, while $\alpha(u)$ and $\alpha(v)$ are the normalization coefficients.

In the frequency domain conversion process, the original image is transformed into 64 vectors, which represent the information distribution of the image on different frequency components. In order to facilitate subsequent neural network processing, these vectors are sampled and their lengths are uniformly fixed to 250.

Afterwards, these 64 vectors with a length of 250 are combined into a 64×250 matrix as input and passed to a specially designed series of CNN networks. These CNN networks gradually extract deep-level information from frequency domain features through structures such as convolutional layers and pooling layers, effectively learning and integrating local and global features in the image frequency domain. Finally, through a fully connected network and activation function processing, the frequency domain features of the image are obtained. Define the input data of the convolutional layer as X_F , the convolution kernel as W , the bias as b_f , and the convolution operation can be demonstrated as:

$$Z = \sum_{i=0}^{k-1} \sum_{j=0}^{k-1} W(i, j) \cdot X_F(p + i, q + j) + b_f \quad (9)$$

where k represents the size of the convolution kernel, and p, q represent the position of the convolution kernel when sliding on the input feature map. After the convolution operation, the data will undergo batch normalization processing, and the formula is:

$$\hat{x}_i = \frac{x_i - E[x_i]}{\sqrt{\text{Var}[x_i] + \epsilon}} \gamma_F + \beta_F \quad (10)$$

Among them, x_i is the value of the input data, and the output value $\hat{x}_i$ is obtained after batch normalization. $E[x_i]$ and $\text{Var}[x_i]$ are the mean and variance of the input data, respectively. To avoid the situation where the denominator is zero, a very small constant ϵ is introduced, and γ_F and β_F are learnable parameters. The data

H_3 obtained after the convolution, batch normalization, and ReLU activation function operations mentioned above is used as the input for the fully connected layer, represented as:

$$\Omega_F = \text{ReLU}(W_F \cdot H_3 + b_F) \quad (11)$$

where $W_F \in \mathbb{R}^{d \times 1}$ is the fully connected layer weight matrix, and b_F is the bias term. This frequency domain feature can capture subtle changes and feature patterns in the image in the frequency domain, complementing spatial domain features and providing a more comprehensive representation of image information for multimodal false information detection.

3.2 Feature fusion module

3.2.1 Co-Attention mechanism module. We adopt the same Co-Attention mechanism (CA) as Jin et al. [18] to enhance the information exchange between different modal features. The CA mechanism achieves dynamic correlation of cross-modal information using the features of one modality as Q and the features of another modality as K and V.

3.2.2 Knowledge-guided priority fusion strategy. In the process of multimodal feature fusion, a single feature interaction is difficult to fully capture cross-modal information. Therefore, we adopt a hierarchical progressive fusion strategy, gradually introducing different modal features in the order of text, image spatial domain, and image frequency domain based on the importance of modal information.

For the fusion of the first-layer text and image spatial domain modalities, two collaborative attention mechanisms CA modules are used to achieve feature fusion between modalities. Specifically, in the first CA module, the main objective is to perform initial fusion between the text feature Ω_T and the image spatial feature Ω_I . Firstly, the text modal features are inputted as Q, and the image spatial modal features are inputted as K and V. After passing through multiple head attention layers and feedforward fully connected neural network layers, residual connections and layer normalization are performed after each layer, as shown below:

$$\Omega_{C_{T \leftarrow I}} = \Omega_T + \text{MHA}(\Omega_T, \Omega_I, \Omega_I) \quad (12)$$

$$\Omega_{C'_{T \leftarrow I}} = \Omega_{C_{T \leftarrow I}} + \text{FFN}(\Omega_{C_{T \leftarrow I}}) \quad (13)$$

where $\Omega_{C'_{T \leftarrow I}}$, as a d-dimensional vector belonging to the real space $\mathbb{R}^d$, represents the visual attention aggregation feature generated based on text feature conditions. In order to achieve a more balanced fusion between the spatial modal features of text and image, a second CA parallel module is introduced. Firstly, the spatial modal features of the image are inputted as Q, and the spatial modal features of the text are inputted as K and V. Then, the calculation operation is performed as in the first CA module, as shown below:

$$\Omega_{C_{I \leftarrow T}} = \Omega_I + \text{MHA}(\Omega_I, \Omega_T, \Omega_T) \quad (14)$$

$$\Omega_{C'_{I \leftarrow T}} = \Omega_{C_{I \leftarrow T}} + \text{FFN}(\Omega_{C_{I \leftarrow T}}) \quad (15)$$

Among them, $\Omega_{C'_{I \leftarrow T}}$ is also located in the real space $\mathbb{R}^d$, with a dimension of d, and is a text attention aggregation feature obtained based on visual spatial conditions. This dual collaborative attention mechanism computing structure enables the model to

optimize specifically for information interaction between different modalities, unlike traditional Transformers that focus on processing information within a single modality. Then concatenate the outputs of the two CA modules together, as shown below:

$$\Omega_C^{(1)} = (\Omega_{C'_{T \leftarrow I}} \oplus \Omega_{C'_{I \leftarrow T}}) W_C^{(1)} \quad (16)$$

Among them, $W_C^{(1)}$ is the projection matrix of the first layer, with a dimension of $2d \times d$, located in the real space $\mathbb{R}^{2d \times d}$. It is worth noting that $\Omega_C^{(1)}$ needs to undergo a specific dimension transformation operation before being input to the next CA layer, transforming it into a $d \times 1$ -dimensional representation.

For the second layer, based on the fusion of text and image spatial features, the frequency domain features of the image are further introduced, and the CA mechanism is used to interact $\Omega_C^{(1)}$ with the frequency domain feature Ω_F , ultimately obtaining the comprehensive fusion feature $\Omega_C^{(2)}$. The entire fusion process is similar to the first layer and can be expressed as:

$$\Omega_{C_{C1 \leftarrow F}} = \Omega_C^{(1)} + \text{MHA}(\Omega_C^{(1)}, \Omega_F, \Omega_F) \quad (17)$$

$$\Omega_{C'_{C1 \leftarrow F}} = \Omega_{C_{C1 \leftarrow F}} + \text{FFN}(\Omega_{C_{C1 \leftarrow F}}) \quad (18)$$

$$\Omega_{C_{F \leftarrow C1}} = \Omega_F + \text{MHA}(\Omega_F, \Omega_C^{(1)}, \Omega_C^{(1)}) \quad (19)$$

$$\Omega_{C'_{F \leftarrow C1}} = \Omega_{C_{F \leftarrow C1}} + \text{FFN}(\Omega_{C_{F \leftarrow C1}}) \quad (20)$$

$$\Omega_C^{(2)} = (\Omega_{C'_{C1 \leftarrow F}} \oplus \Omega_{C'_{F \leftarrow C1}}) W_C^{(2)} \quad (21)$$

To fully leverage the contextual information from both fusion stages, we concatenate the features from the first fusion layer, $\Omega_C^{(1)}$, with those from the second layer, $\Omega_C^{(2)}$. This combined feature vector is then passed through a final linear projection to produce the ultimate feature representation Ω_{Final} , which is used for the downstream classification task. This can be formulated as:

$$\Omega_{Final} = (\Omega_C^{(1)} \oplus \Omega_C^{(2)}) W_{Final} \quad (22)$$

3.3 Fake news detection module

Given $f_1 = \Omega_{Final}$, the variable framework based on collaborative attention fusion features constructed in this article ultimately outputs the probability of a graphic and textual tweet being false information. The calculation formula is as follows:

$$\hat{y}_1 = \text{softmax}(\text{ReLU}(f_1 W_{f1}) W_{s1}) \quad (23)$$

where W_{f1} refers to the parameters of the fully connected layer, and W_{s1} is the parameter of the linear layer in the softmax layer. The specific expression of the loss function is:

$$L(\Theta) = -y_1 \log(\hat{y}_1) - (1 - y_1) \log(1 - \hat{y}_1) \quad (24)$$

where y_1 represents the true label. Θ covers all learnable parameters in SVPF-Net, which will be continuously optimized and adjusted during the model training process to improve its performance and prediction accuracy.

Table 1: Comparison of the performance among various models on the Weibo and PHEME datasets. The best performance for each metric is highlighted in bold.

Dataset	Methods	Accuracy	Fake News			Real News		
			Precision	Recall	F1	Precision	Recall	F1
Weibo	Text	0.876	0.885	0.871	0.878	0.865	0.878	0.871
	Spatial	0.857	0.850	0.877	0.863	0.863	0.834	0.848
	Freq	0.717	0.728	0.724	0.726	0.706	0.710	0.708
	NeuralTalk [39]	0.726	0.794	0.713	0.692	0.684	0.840	0.754
	att-RNN [18]	0.772	0.854	0.656	0.742	0.720	0.889	0.795
	EANN [41]	0.782	0.827	0.697	0.756	0.752	0.863	0.804
	TextGCN [45]	0.787	0.975	0.573	0.727	0.712	0.985	0.827
	MVAE [20]	0.824	0.854	0.769	0.809	0.802	0.875	0.837
	SpotFake [37]	0.869	0.877	0.859	0.868	0.861	0.879	0.870
	HMCAN [34]	0.885	0.920	0.845	0.881	0.856	0.926	0.890
	MCAN [42]	0.899	0.913	0.899	0.901	0.884	0.909	0.897
	CAFE [4]	0.840	0.855	0.830	0.842	0.825	0.851	0.837
	BMR [46]	0.889	0.874	0.894	0.884	0.904	0.885	0.895
	MVML [6]	0.910	0.915	0.871	0.893	0.906	0.939	0.922
	SVPF-Net (ours)	0.912	0.892	0.930	0.911	0.931	0.894	0.912
PHEME	SAFE [49]	0.811	0.827	0.559	0.667	0.806	0.940	0.866
	att-RNN [18]	0.850	0.791	0.749	0.770	0.876	0.899	0.888
	EANN [41]	0.681	0.685	0.664	0.694	0.701	0.750	0.747
	TextGCN [45]	0.828	0.775	0.735	0.737	0.827	0.828	0.828
	MVAE [20]	0.852	0.806	0.719	0.760	0.871	0.917	0.893
	SpotFake [37]	0.823	0.743	0.745	0.744	0.864	0.863	0.863
	HMCAN [34]	0.881	0.830	0.838	0.834	0.910	0.905	0.970
	MVML [6]	0.893	0.826	0.751	0.781	0.917	0.941	0.928
	SVPF-Net (ours)	0.915	0.857	0.843	0.850	0.936	0.942	0.939

4 Experiments

4.1 Experimental Setup

This subsection introduces the datasets, comparison methods, hyperparameter settings, and evaluation metrics.

4.1.1 Datasets. We focus on multimodal fake information detection, mainly using text and image data. We adopted two commonly used real-world datasets, one from the Chinese Weibo dataset [18] and the other from the English PHEME dataset [51], to verify the performance of SVPF-Net. They correspond to fake news detection tasks in both Chinese and English text scenarios. After preprocessing, tweets containing videos or lacking text and images were removed, and the baseline testing data partitioning scheme was used.

4.1.2 Baselines. To comprehensively verify the effectiveness of the proposed method, we selected representative baseline models for comparative experiments. For the Weibo dataset, we selected three unimodal models, including Text, Spatial, and Freq, which are based on the MCAN [42] model framework and only retain the unimodal results obtained from the text modality, image spatial modality, and image frequency domain modality feature information, respectively. Furthermore, we also introduce a representative unimodal graph neural network model, TextGCN [45]. Regarding multimodal baselines, we have selected several representative models from recent years, including NeuralTalk [39], att-RNN [18], EANN [41], MVAE [20], SpotFake [37], HMCAN [34], MCAN [42],

CAFE [4], BMR [46] and MVML [6]. They cover various fusion strategies such as feature layer fusion, decision layer fusion, and attention collaborative fusion to compare the superiority of the fusion strategy proposed in this paper. For the PHEME dataset, select models that are clearly based on the PHEME dataset and can obtain complete performance indicators from the 14 models mentioned above as comparison objects to ensure the reliability and comparability of experimental results.

4.1.3 Parameter Settings. We set the maximum text length to 160 for the Weibo dataset and 60 for the PHEME dataset. The embedding dimension model_dim for each modality is set to 256, the number of layers m for the Cross-Attention (CA) layer is 2, and the ffn_dim is 2048. The hidden layer dimension in the classification head is 16. For the Weibo dataset, we load the RoBERTa_zh_L12 model to acquire basic understanding of Chinese corpora; for the PHEME dataset, we load the RoBERTa-base model pre-trained on large-scale English corpora. The Adabelief optimizer and Adam optimizer are used for the Weibo and PHEME datasets, respectively. The learning rate and weight_decay are adjusted according to the number of trainable parameters. We train the model for 50 epochs on the Weibo dataset and 100 epochs on the PHEME dataset, with an early stopping mechanism adopted to prevent overfitting during model training.

4.1.4 Evaluation Metrics. Fake news detection is essentially a binary classification task. We selected four classification evaluation metrics, namely Accuracy, Precision, Recall and F1-Score. These

Table 2: Ablation study of different modules on the Weibo and PHEME datasets. The performance of the full model is highlighted in bold.

Dataset	Modules					Accuracy	F1	Fake News			Real News		
	Text	Image (CNN)	Image (Trans)	DCT	CA			Precision	Recall	F1	Precision	Recall	F1
Weibo	✓	✓	✓	✓	✓	0.912	0.912	0.893	0.930	0.911	0.932	0.895	0.913
	✗	✓	✓	✓	✓	0.747	0.746	0.750	0.717	0.733	0.744	0.775	0.759
	✓	✗	✓	✓	✓	0.902	0.902	0.945	0.879	0.911	0.914	0.895	0.904
	✓	✓	✗	✓	✓	0.909	0.909	0.893	0.922	0.908	0.925	0.897	0.910
	✓	✗	✗	✓	✓	0.901	0.900	0.940	0.850	0.893	0.870	0.949	0.908
	✓	✓	✓	✗	✓	0.906	0.906	0.894	0.914	0.904	0.917	0.898	0.908
	✓	✓	✓	✓	✗	0.892	0.892	0.874	0.908	0.891	0.910	0.877	0.893
PHEME	✓	✓	✓	✓	✓	0.915	0.895	0.857	0.844	0.850	0.937	0.943	0.940
	✗	✓	✓	✓	✓	0.727	0.660	0.525	0.500	0.512	0.800	0.815	0.808
	✓	✗	✓	✓	✓	0.896	0.872	0.836	0.797	0.816	0.919	0.936	0.927
	✓	✓	✗	✓	✓	0.904	0.890	0.787	0.922	0.849	0.966	0.898	0.931
	✓	✗	✗	✓	✓	0.899	0.879	0.828	0.828	0.828	0.930	0.930	0.930
	✓	✓	✓	✗	✓	0.896	0.878	0.781	0.891	0.832	0.953	0.898	0.925
	✓	✓	✓	✓	✗	0.874	0.851	0.750	0.844	0.794	0.933	0.885	0.908

diverse evaluation metrics are used to comprehensively assess the performance of the proposed model in this work, ensuring the accuracy and comprehensiveness of the evaluation results.

4.2 Performance Comparison

To validate the effectiveness and superiority of our proposed model, SVPF-Net, we conducted comprehensive comparative experiments with several baseline methods on the Weibo and PHEME datasets. The detailed results are presented in Table 1. The analysis shows that our model achieves state-of-the-art performance on both datasets. Specifically, on the Weibo dataset, SVPF-Net achieved an accuracy of 0.912 and a fake news F1-score of 0.911. Notably, its recall reached 0.930, consistently surpassing strong baselines such as MCAN and MVML. This performance advantage is even more pronounced on the PHEME dataset, where our model delivered a top accuracy of 0.915 and an F1-score of 0.850. Compared to the second-best model, MVML, SVPF-Net improved the accuracy by 2.4%. These consistent and significant improvements across diverse datasets validate that the proposed semantic visual enhancement and modality-priority progressive fusion strategy are highly effective for multimodal fake news detection.

4.3 Ablation Study

4.3.1 Component Contribution Analysis. To assess each component’s contribution, we conducted an ablation study, with results presented in Table 2. The analysis confirms that each module is indispensable for the model’s final performance. The text modality acts as the semantic backbone, as its removal leads to the most significant degradation, causing the accuracy to drop by 18% on Weibo and 20.5% on PHEME. Our dual-branch visual feature extraction, which combines a CNN for local traces and a Transformer for global context, is also validated; removing either component demonstrably harms performance, confirming their complementary nature in building a comprehensive visual representation. Most critically, the Cross-Attention (CA) fusion mechanism proves to be

essential. Disabling this module results in a severe performance decline, with accuracy dropping by 2.1% on Weibo and 4.4% on PHEME. This demonstrates that beyond powerful feature extraction, an effective mechanism to model intricate cross-modal interactions is paramount to the model’s success.

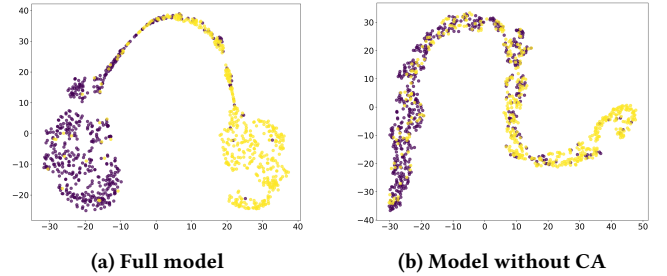


Figure 2: t-SNE visualizations of multimodal embeddings on the Weibo dataset. (a) shows the embedding distribution from the full SVPF-Net model, while (b) illustrates the distribution when the Cross-Attention (CA) module is ablated. Different colors represent different classes (fake/real news).

4.3.2 t-SNE Visualization of Embeddings. To intuitively demonstrate the impact of the Cross-Attention (CA) module, we visualized the multimodal embeddings on the Weibo dataset using t-SNE, as shown in Figure 2. Figure 2a displays the embeddings from the full SVPF-Net model, revealing distinct and well-separated clusters for fake and real news, indicating strong discriminative power. In contrast, Figure 2b, which visualizes embeddings when the CA module is removed, shows significantly more overlap and less clear separation between the two classes. This visual evidence further supports our ablation study findings in Table 2, highlighting the critical role of the CA module in achieving effective cross-modal interaction and generating highly distinguishable representations for fake news detection.

Table 3: Comparison of different fusion methods. $\Omega_C^{(1)}$ represents the fused features of text and image spatial domain, while $\Omega_C^{(2)}$ adds the fusion of image frequency domain features. The methods evaluated are Concatenation Fusion (concatenating $\Omega_C^{(1)}$ and $\Omega_C^{(2)}$), Add Fusion ($\Omega_C^{(1)} + \Omega_C^{(2)}$), and an ablation study using only the subsequent features ($\Omega_C^{(2)}$ only).

Dataset	Fusion Method	Accuracy	F1-Score
Weibo	Concatenation Fusion	0.912	0.912
	Add Fusion	0.902	0.902
	Ablation	0.908	0.908
PHEME	Concatenation Fusion	0.915	0.895
	Add Fusion	0.906	0.887
	Ablation	0.911	0.894

Table 4: Performance comparison of different multimodal fusion orders. T, V, and F represent the Text, Visual, and Frequency-domain modalities, respectively. The fusion order indicates the sequence in which the modalities are integrated.

Dataset	Fusion Order	Accuracy	F1-score
Weibo	T $\rightarrow$ V $\rightarrow$ F	0.912	0.912
	T $\rightarrow$ F $\rightarrow$ V	0.903	0.903
	V $\rightarrow$ F $\rightarrow$ T	0.899	0.899
PHEME	T $\rightarrow$ V $\rightarrow$ F	0.915	0.895
	T $\rightarrow$ F $\rightarrow$ V	0.901	0.882
	V $\rightarrow$ F $\rightarrow$ T	0.901	0.880

4.3.3 Analysis of Fusion Method. To validate our choice of fusion strategy, we compared our model’s concatenation fusion with a simpler add fusion method, as detailed in Table 3. The results demonstrate that concatenation fusion consistently outperforms the alternative on both datasets, achieving the highest accuracy of 0.912 on Weibo and 0.915 on PHEME. This indicates that concatenating feature vectors from different fusion stages is more effective at preserving their distinct and complementary information, leading to a more robust final representation for the detection task.

4.3.4 Analysis of Fusion Order. Furthermore, we explored the impact of the fusion order for the different modalities, as shown in Table 4. We evaluated three distinct sequences for integrating the features of the Text (T), Visual (V), and Frequency-domain (F). The results unequivocally show that the T $\rightarrow$ V $\rightarrow$ F sequence, which is central to our modality-priority fusion strategy, yields the best performance on both datasets, achieving a top accuracy of 0.912 on Weibo and 0.915 on PHEME. This strongly validates our hypothesis that initializing the fusion process with the semantically rich text modality provides a robust context that most effectively guides the subsequent integration of visual information.

4.4 Discussion

The superior performance of our model, validated across both datasets, is attributable to two principal innovations: the semantic

visual enhancement module and the knowledge-guided priority fusion strategy. The ablation results presented in Table 2 confirm that our dual-branch visual extractor, which synergizes a CNN’s ability to capture local manipulation artifacts with a Transformer’s capacity for global semantic reasoning, creates a far more comprehensive visual representation than either architecture alone. This enhanced visual understanding is then effectively integrated using our modality-priority fusion approach. As demonstrated by the fusion order study in Table 4, the success of the Text $\rightarrow$ Visual $\rightarrow$ Frequency sequence validates our hypothesis that using the text modality as a semantic anchor provides crucial context, guiding the subsequent fusion of visual features. The CA mechanism facilitates this hierarchical integration, enabling a deeper cross-modal interaction that results in more discriminative features, as visually evidenced by the clear class separation in Figure 2. This moves beyond simple feature concatenation to a more effective, context-aware fusion process.

5 Conclusion

In this paper, we address the critical challenges of inadequate visual representation and inefficient modality fusion in multimodal fake news detection. To this end, we propose a novel detection method, SVPF-Net, centered on semantic-driven visual enhancement and a knowledge-aided modality-priority fusion strategy. Our core contributions are twofold: (1) a semantic visual enhancement module that synergistically combines the local-feature extraction power of CNNs with the global context modeling of Transformers, yielding a rich and structured visual representation that captures interpretable semantic cues; and (2) a knowledge-guided priority fusion strategy that leverages a hierarchical Co-Attention mechanism to realize progressive modality fusion. Specifically, this strategy takes textual features as a semantic anchor, capitalizing on the strong semantic consistency of text to guide the selective integration of visual information, effectively mitigating inter-modal interaction inefficiencies and semantic fusion imbalances. Extensive experiments on the real-world datasets, including Weibo and PHEME demonstrate that SVPF-Net significantly outperforms existing state-of-the-art methods. The results validate SVPF-Net’s ability to effectively model complex cross-modal forgery patterns, thereby providing a more accurate and robust solution for ensuring the trustworthiness of the Web ecosystem.

In future work, we plan to incorporate a broader range of heterogeneous signals, such as video sequences and social network propagation graphs, to construct a more comprehensive and holistic detection framework. Furthermore, we will explore model compression and lightweight architectural designs, aiming to significantly reduce computational overhead and resource consumption while maintaining high detection accuracy.

Acknowledgments

This work was supported by the National Natural Science Foundation of China (Grant No. 62276101 and No. 72204037) and the Natural Science Foundation of Guangdong Province, China (Grant No. 2024A1515510014).

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Appendix A: Detailed Specifications of Baseline Models

To validate the performance of SVPF-Net, we compare it against a wide range of state-of-the-art baselines. These models are categorized based on their architectural designs and modality processing strategies.

A.1 Unimodal Baselines

To evaluate the contribution of individual modalities and the effectiveness of graph-based text modeling, we consider the following unimodal baselines:

- **Text, Spatial, and Freq:** These are unimodal variants derived from the MCAN framework [42] used for ablation study, retaining only the textual branch, spatial visual branch, or frequency-domain visual branch, respectively.
- **TextGCN** [45]: A representative graph-based text classification model that builds a heterogeneous corpus graph with word and document nodes. It utilizes Graph Convolutional Networks (GCN) to capture global word co-occurrence and learn discriminative embeddings.

A.2 Multimodal Baselines

The following models represent diverse fusion paradigms, ranging from early generative approaches to modern cross-modal attention networks:

- (1) **NeuralTalk** [39]: A generative model based on a deep recurrent architecture that employs a CNN for visual feature extraction and an LSTM for natural language generation, mapping images directly to textual descriptions.
- (2) **att-RNN** [18]: Utilizes an LSTM to fuse textual and social-context features with image features extracted by VGG-19, while employing an attention mechanism for dynamic visual-text alignment.
- (3) **EANN** [41]: Extracts textual features via Text-CNN and visual features via VGG-19, jointly training a fake news detector to learn event-invariant representations.
- (4) **MVAE** [20]: Combines Bi-LSTM text features and VGG-19 visual features; learns shared cross-modal latent representations through joint optimization of reconstruction and KL-divergence losses.
- (5) **SpotFake** [37]: Employs BERT for text and VGG-19 for images, concatenating both feature vectors for fake news classification via fully connected layers.
- (6) **HMCAN** [34]: Uses BERT to extract textual features and ResNet-50 to extract regional visual features. A hierarchical multimodal context attention network fuses intra- and inter-modal relations, followed by hierarchical encoding for text semantics and final classification.
- (7) **MCAN** [42]: Extracts spatial features using VGG-19, frequency-domain features via CNN, and textual features via BERT. Multi-layer co-attention modules progressively fuse visual and textual representations to capture inter-modal dependencies.
- (8) **CAFE** [4]: Extracts textual features via BERT and visual features via ResNet, performs cross-modal alignment and ambiguity learning using KL divergence, and adaptively aggregates uni- and cross-modal representations for classification.
- (9) **BMR** [46]: Integrates an improved mixture-of-experts mechanism to fuse tri-view representations (text, image modality, and image semantics), performing disentanglement and reweighting through consistency learning to guide fake news detection.
- (10) **SAFE** [49]: The method employs Text-CNN to extract textual features and utilizes a modified VGG-19 for visual feature extraction. Subsequently, the cosine similarity between the textual and visual features is calculated to quantify the cross-modal consistency of the news content.
- (11) **MVML** [6]: The method first employs BERT and ResNet/Faster R-CNN to extract features from text, image scenes, and objects, respectively. Subsequently, it captures cross-modal semantic relationships from both global and local perspectives by constructing text-scene and text-object heterogeneous graphs and applying graph attention-based reasoning. Finally, the model utilizes a cross-view attention mechanism for feature fusion and incorporates a mutual learning mechanism based on KL divergence.